Trust Mechanism Fuzzy Rules Intelligent Car Real-Time Diagnostic System

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***Abstract*—In this paper we propose a real-time fuzzy logic system composed as early failure diagnostic of the car driveline. The IoT infrastructure of the system was designed to prevent interference with the original car layouts. We propose a new model which defines a devoted trust mechanism supporting ad-hoc knowledge exchange between cooperating fuzzy systems run on each connected car. The paper presents electronic IoT infrastructure as well as theoretical and practical foundations for the developed fuzzy logic control module. The system has been tested and verified based on practical experiments.**

***Index Terms*—Smart car, IoT, fuzzy logic, trust mechanism**

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I. Introduction

The development of modern information technologies that allow us to combine widely available elements of the IoT infrastructure with artificial intelligence methods allows us to create very interesting and useful solutions. Hardware platforms provide us with a number of sensors and possibility of wireless communication, thanks to which it is possible to use them in cars without interfering with their integral systems. In addition, the use of Rasberry Pi class computers enables the creation of inexpensive solutions equipped with artificial intelligence methods operating in real time during the operation of the vehicle. Artificial intelligence methods in recent years are often based on the innovative Federated Learning technique [1]. The learning process takes place on distributed data without the need to transfer them to a central server. In this type of approach, only the effects of local learning that take place in distributed local servers are sent to the central server. This process is aggregated and affects the update of the global model. Federated learning applies to many different aspects, including medical data [2], [3]. Federated learning requires a certain level of security and confidentiality, which is quite a challenge [4]. The exchange of data between local servers and the central server related to Federated Learning requires the use of authentication and authorization. This mechanism is called Trust Mechanism. In our work, we will propose a trust mechanism based on the similarity measure of fuzzy sets [5]. The system we propose is based on IoT solutions and its overriding goal is to provide early warning of an emerging fault, before the effects of this fault are perceptually detected by the user. This solution is particularly important in the case of vintage cars, where access to parts is limited or absent and their repair is expensive. Due to the fact that we have many different car brands, different models, for each of these vehicles the fuzzy system will be proposed to tune individual vehicles and periodically update the central model in the server application.

There are many different applications of modern vehicle information technology methods. Diagnostic systems for smart vehicles operate on various features. The model proposed in [6] presented an algorithm for front detection where model of Singular Spectrum Decomposition (SSD) analyzed the input data. Monitoring systems are very often working in 5G mode [7]. In the development of AIoT also depends on electronics and sensors which compose IoT infrastructures providing data for applied AI models. A system showed in [8] presented application of SPIN-V sensors in smart parking system. Very often camera vision is used to support drivers for more efficient car control [9]. Li-Fi and ultrasonic sensor were used in [10] to design collision prevention system. Operating systems which hold developed applications are mainly based on Android platforms or mobile computing. The model proposed in [11] has presented a smartphone flush measurement for car body assembly. An application for mobile devices was also examined for helping in parking by using rfid approach [12]. In [13] was examined multi level system used for smart metering in which data flow structure was composed of co-working software modules. The model proposed in [14] was developed for 5G mode where all elements cooperate on networking. Discussion on next generation technologies for smart car systems was presented in [15]. An analysis of potential developments in the field of applied smart car technologies was presented in [16]. In [17] was presented a discussion on wireless technologies which serve as data transfers for AI models.

AI models developed for smart cars working in IoT mode are mainly based on fuzzy systems, since such models are easy to tune and compose if we have an expert knowledge. There are also very efficient and precise in various applications in vehicle diagnostics. In [18] was proposed Mamdani fuzzy interference based model for car parking support. Fuzzy interference model was also applied in [19] to automatics measurement in engineering. A wind turbine engine elements were examined by fuzzy control model in [20], as a result the system was able to predict condition using proposed assembled multidimensional functions. Also in railway applications fuzzy control system was used to predict maintenance and usage of

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parts. It is also very important to support driving methodology as proposed in [21]. The fuzzy model proposed in [22] was designed to control aeroservoelastic characteristic of reusable vehicles, while the model presented in [23] was using IoT and fuzzy model for condition monitoring of a train. Several car control systems work in hybrid mode, where AI models cooperate with analytical predictive approaches [24].

In particular, the modern possibilities provided by the development of information technology and artificial intelligence are interesting for owners of old classic cars. In case of these often unique vehicles, it is very important to detect potential failures early in order to limit the damage resulting from the progressive degradation of certain elements. In many cases, it is no longer possible to purchase original not used parts, and in extreme situations, they must be recreated or regenerated in specialized companies, which is associated with very high repair costs. The AIoT system proposed by us is responsible for monitoring operation of drive engine and cooling systems in real time. As part of the research work, the system was installed in the 1991 Cadillac Seville test car, shown in Fig. 2. In our solution, the graphical user interface is an application implemented by us and installed on a radio receiver with the Android system.

II. Project goal

In most cars, built-in diagnostic tools protect the driveline against overheating and inform the driver only about a failure or imminent risk of failure as a standard. With a view to increase service life of the vehicle and early prevent breakdowns, a system based on fuzzy logic was created. Based on the temperature measurements of the car drive system, taking into account the load to which the system is subjected, conditions of early danger can be detected. Examples of situations affecting degradation of the drive system may be deteriorating cooling conditions resulting from increasing contamination of the radiator, decreasing level of the coolant fluid. In the event of contamination of the cooling system, monitoring systems are not used, while in the case of too low level of coolant, solution is monitored from threshold level of the liquid. Therefore, based on numerous observations and registration of certain parameters, it was decided to create a system aimed at early detection of a potential increase in the risk of damage. The purpose of this procedure is to report the need for in-depth diagnostics of cooling and drive systems. This type of monitoring, in particular, can be widely used in classic cars due to their poor equipment with advanced control systems and difficult availability of spare parts. As a result, the use of additional diagnostic system will positively affect the extension of the life of car. In order to obtain the usefulness of the diagnostic system, it is assumed that it will be created based on the IoT infrastructure, which will minimize extent interfere with original components of the diagnosed vehicle. Based on numerous measurements and observations, it was decided to create a fuzzy system. The motivation for creating such a system was specificity of measurement data, which were naturally fuzzy features: temperature, drive system load, engine rotational speed. The tests were carried out for a gasoline engine cooperating with an automatic gearbox and a diagnostic system was proposed for such a solution.

III. Description of the system concept

The basic assumption in the implementation of the measurement infrastructure was not to interfere with the systems installed in the vehicle. The basic sensors built into the system were the DS18B20 temperature sensor. All these sensors are mounted on the outer side of metal conductors that conduct heat well and are tightly isolated at the installation site from the environment. Such assembly ensured obtaining a temperature measurements close to that prevailing in the medium inside the conduit. Such assembly of sensors allows us to perform measurement for the needs of our system and does not change the structure of the drive system. Thanks to which the initial state of the drive system can be restored at any time by disassembling the sensors installed externally. WiFi communication ensures the transmission of measurement data between individual sensors and the superior server application. For this purpose, each of the installed temperature sensors was connected to an electronic system ensuring WiFi communication - ESP8266 system.

The system is also based on measuring the speed of the vehicle. This value is determined based on the vehicle speed sensor installed as standard in the car. The speed sensor located in the gearbox was connected to the measuring system with an additional connector. The signal representing the rotational speed of the engine shaft was obtained in a similar way. The place of obtaining this signal was the ignition control module located in the ignition apparatus.

In addition, studies have shown that the way the automatic transmission works, it also has an impact on the temperature of the drive system. In the case of operation with the use of a torque converter in the gearbox, there are greater losses of energy released in the form of heat than in the case of operation with the use of a viscous coupling. For this reason, an additional measuring system was made to record the mode in which the gearbox works and what is the currently selected gear.

The main server application responsible for collecting data from sensors and their ongoing analysis was located in a radio receiver equipped with the Android operating system. The radio receiver is additionally responsible for broadcasting the WiFi network within the vehicle, thanks to which it is possible to communicate the electronic systems of each sensor with the server application. The ESP8266 systems were programmed so that they could send measurement data to the Rest API installed in the radio receiver.

The trust mechanism used in federated learning was created based on the similarity measure of fuzzy sets. Federated learning is based on the fact that the server application sends the knowledge developed by the system to all subordinate agents in the form of individual cars in the form of fuzzy sets describing all measurement quantities. Based on them, each of the measured quantities in the first stage of the fuzzy reasoning system is blurred. During these operations, individual agents fine-tune the knowledge obtained from the server within their

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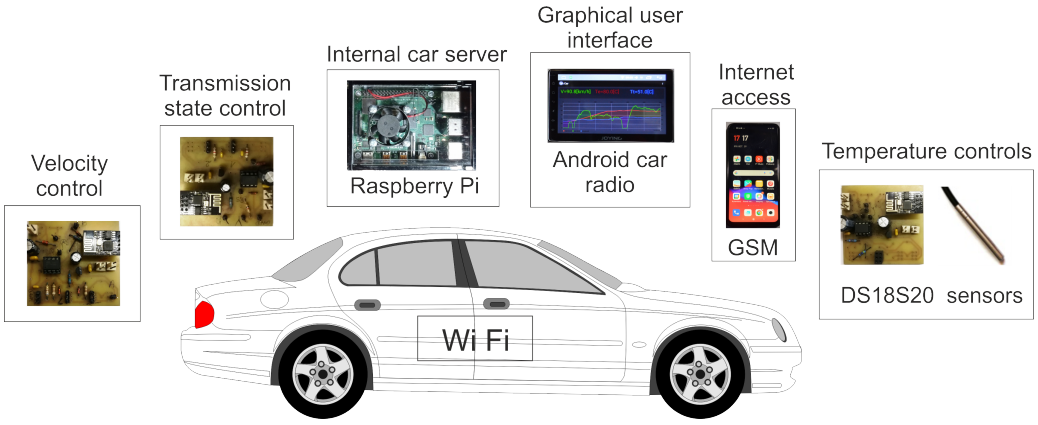


Fig. 1. Diagram of the IoT infrastructure in the car to obtain data for analysis. In the model we have implemented Android control unit and developed 4 control electronic modules which are implemented in the car to collect data and send it via gsm model to our data server for knowledge distillation and fuzzy rules composition.



Fig. 2. Car used for our research project is Cadillac Seville year 1991.

operation. Periodically, after traveling a certain number of kilometers, each agent sends its modified knowledge in the form of fine-tuned fuzzy sets to the server application. The server application is equipped with a trust mechanism module whose task is to determine whether the knowledge sent by the agent is valuable and whether we are dealing here with an attempt to send false data. The superior tool here is the measure of similarity of fuzzy sets. The system assumes that corrections made by a single agent cannot differ significantly from the general data generated by the server. Therefore, the trust mechanism for each of the sent thugs calculates the measure of its similarity to the template on the basis of which it was created. If the average similarity of each of the sets describing a given variable is large enough (above the set threshold), the trust mechanism accepts the received data and uses them in the process of teaching server applications. If the average similarity measure of any of the quantities is too small, the system does not use any of the data sent in a given packet.

We will now define what the similarity function is for fuzzy sets. Let $\mathbb{F}$ be the class of all fuzzy sets with membership functions $\mu : X- > [0,1]$.

Let $A \in \mathbb{F}$, $B \in \mathbb{F}$, $C \in \mathbb{F}$ then the function $S : \mathbb{F}^2 \to [0,1]$ is meeting the following assumptions:

1) $S(A,A) = 1$,
2) $S(A,A^C) = 0$,
3) $S(A,B) = S(B,A)$,
4) if $A \subseteq B \subseteq C$ than $S(A,B) \geq S(A,C)$ and $S(C,B) \geq S(A,C)$

In particular, in the operation of our system, we used the measure of similarity of fuzzy sets defined by the formula:

$$S(A,B) = 1 - \left(\frac{\int_0^1 |\mu_A(x) - \mu_B(x)|^\alpha \, dx}{\int_0^1 \mu_A^\alpha(x)dx + \int_0^1 \mu_B^\alpha(x)dx} \right)^{1/\alpha} \quad (1)$$

The operation of the vehicle causes wear of the drive and brake system components, which may lead to the appearance of additional vibrations. These vibrations cause accelerated wear of other elements of the drive system. For example, a worn drive-shaft joint will cause additional vibrations on one side of the vehicle's driveline, which will be so muffled by the vehicle's suspension system that it may not be felt in the car's cabin by the driver. Another case of the usefulness of a vibration measurement system is the determination of its source. In this case, the vibrations may be felt by the driver, but it is impossible to locate their source. Another example of extra vibration is an unevenly worn brake disc. In this case, vibrations will appear mainly when the car is braked. In all of these examples, it is important to detect vibration abnormalities early, before they can be clearly felt by the driver. It should be emphasized that the driver, subjected to slowly increasing stimuli, gradually gets used to them, which affects the delay in detecting any irregularities by his senses. The pseudocode 1 presents the trust mechanism used in our solution based on determining the similarity measure of fuzzy sets.

IV. Fuzzy diagnostics system

In older cars, the internal diagnostics system is not as extensive as in modern cars. Both in modern and classic cars, the diagnostic system informs the driver only about exceeding the permissible maximum engine temperature, regardless of the operating mode and load. This approach to diagnostics does not allow for the observation of the wear of certain

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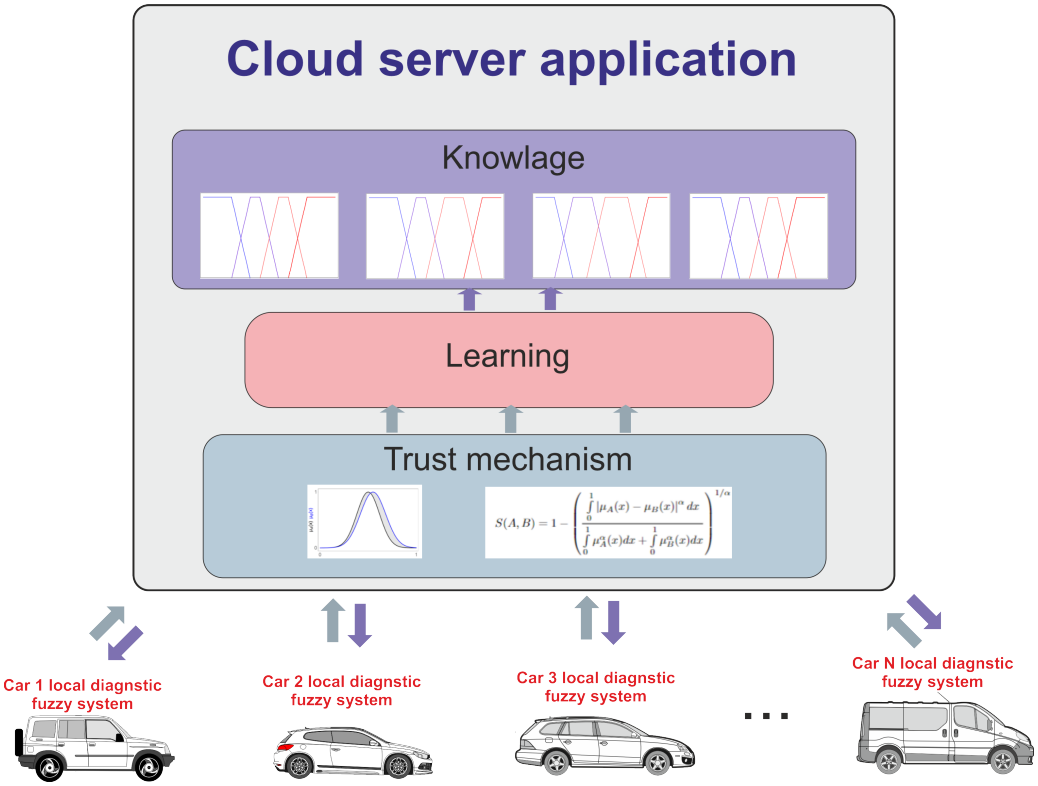


Fig. 3. The diagram shows the concept of how the system work, where single agents - cars learn by tuning the system while driving. After passing it to the server application, which checks what is the measure of similarity between the fuzzy sets sent by the agent compared to the sets of general knowledge. If the difference is too large the system does not accept such correction. Otherwise if it is within the acceptable range, it is transferred to the learning block, which, based on corrections from many different cars, updates the set of basic fuzzy functions. In our system the process is continued while the knowledge is constituted for global improvement.

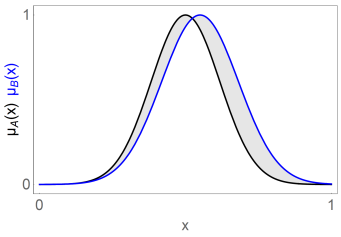


Fig. 4. An example illustrating the measure of similarity of two fuzzy sets with membership functions μ_A and μ_B, respectively, the size of the depends on the shaded area between measures.

components. You can non-invasively monitor temperatures at various measurement points of the drive system, which allow for earlier detection of certain abnormalities in the drive system and may prevent damage or reduce its scope. Fig. 6 shows a diagram of the location of temperature sensors in the car cooling system. This diagram shows the components of the power block (engine, gearbox) and the cooling system. In addition, the places where the temperature sensors are installed are marked. The photo below shows a cooler with symbolically marked places for mounting temperature sensors.

The precise temperature sensors (DS 18B20) installed in accordance with Fig. 6 made it possible to measure the temperature of the medium at important points in the drive system. In every vehicle with a liquid-cooled engine, the temperature at the radiator inlet and outlet can be measured. In Fig. 6 these are sensors T2 and T3. Moreover, the analyzed drive system has an engine oil cooling system, which lubricating the engine oil is supplied to the heat exchanger located on the left side in the radiator shown in Fig. 6, as the measuring points are marked as T4 and T5. At the bottom there is

Algorithm 1: Trust mechanism based on measure of similarity of fuzzy sets

1: Start,
2: Define parameters of the algorithm: $\delta \in [0, 1]$ - trust level.
3: k-agent (car) send data to server (fuzzy sets A_k^i for $i = 1, \ldots, n$)
4: **for all** A_k^i **do**
5: server calculates measure of similarity of fuzzy sets A_k^i and A^i using equation (1),

$$s_k^i = S(A_k^i, A^i)$$

where A_k^i is sent by car and A^i is stored by server
6: **if** $s_k^i < 1 - \delta$ **then**
7: reject data from car k
8: **end if**
9: **end for**
10: accept data from car k and update server knowlage
11: Stop.

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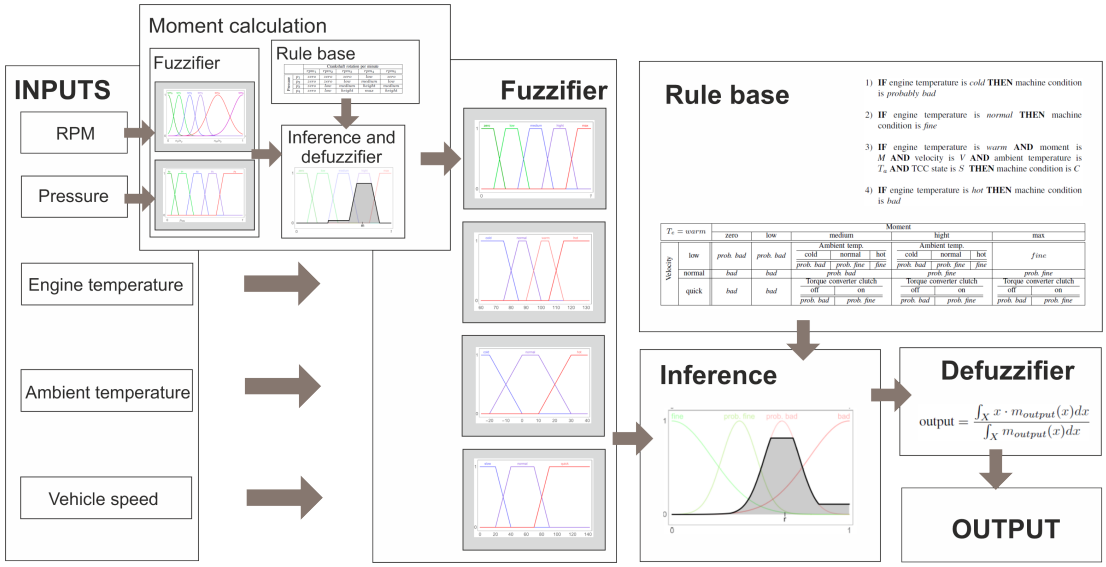


Fig. 5. Diagram of the proposed fuzzy system data flow. Input data readings are processed by proposed fuzzifier. Results are forwarded to rule based model which is using interference caluclations on the way to final decision form the model.

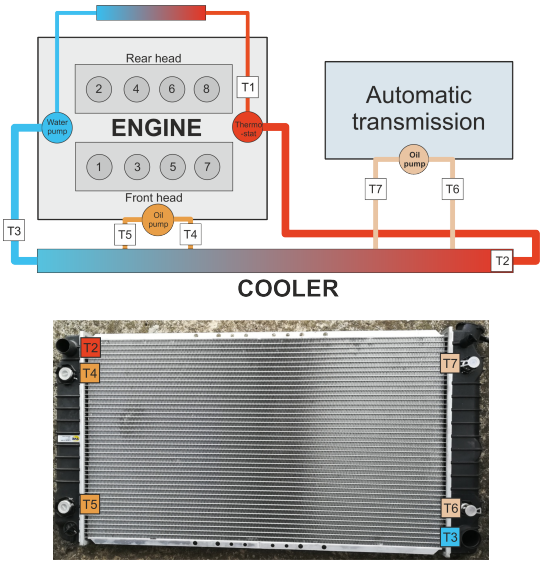


Fig. 6. Location of temperature sensors mounted on the outer part of the pipes of the cooling system. The diagram shows the drive block (engine and gearbox) and the cooling system. The graphic shows the radiator with marked mounting points for the sensors.

a place where the engine oil is fed to the heat exchanger, and at the top there is a place where the oil is taken. This solution is a system with a counter-flow heat exchanger, which allows for better stabilization of the temperature of the engine lubricating oil. The driveline mounted in the test vehicle equipped with an automatic gearbox also has a gearbox oil heat exchanger, which heat exchanger is installed in the right part of the radiator shown in Fig. 6 (measuring points T6 and T7). Measurement of these temperatures and inference based on this data of the condition of the drive system requires additional information such as speed and load of engine and vehicle speed. Vehicle speed has a significant impact in the ”efficiency” of the radiator. The engine speed was measured by using the signal from the ignition module, and the engine load is determined on the basis of the signal from the intake manifold pressure sensor, and the vehicle speed signal was obtained from the speed sensor installed in the gearbox.

For the interpretation of measurement results, a dedicated fuzzy inference system has been developed, as presented in Fig. 3. It works based on the measurements of the engine temperature, ambient temperatures and the determination of the engine load during its operation. This is a proprietary solution at the research stage. Tuning the system is related to the variety of different car models used. In this way, the system develops a knowledge base. In further works, when we obtain more data recorded from various vehicles, we will attempt to automatically tune parameters [25], [26]. The diagnostic system was built as a fuzzy inference system, because it is possible to define the rules corresponding to the correct and disturbed operation of the tested system and the emergency state in a natural way. Takagi-Sugeno model was used. A diagram showing the operation of the created diagnostic system operating in a single car is shown in Fig.5. In the case of this system, it is difficult to set sharp ranges for individual values of the tested parameters, also due to their burden with measurement errors.

The basic and most important assumption for the diagnostic system is to detect the operating states of the drive system, which may indicate certain irregularities in its operation. They are not yet an emergency states. The purpose of this system is to prevent a serious vulnerability which the deteriorating condition of the tested car may lead to.

Developed method of determining the fuzzy value of the engine load torque is constructed for naturally aspirated gasoline engines. The parameters allowing to determine the fuzzy value are:

- engine speed - measured with the measuring system attached to the ignition module installed in the ignition device, this parameter for any engine has specific char-

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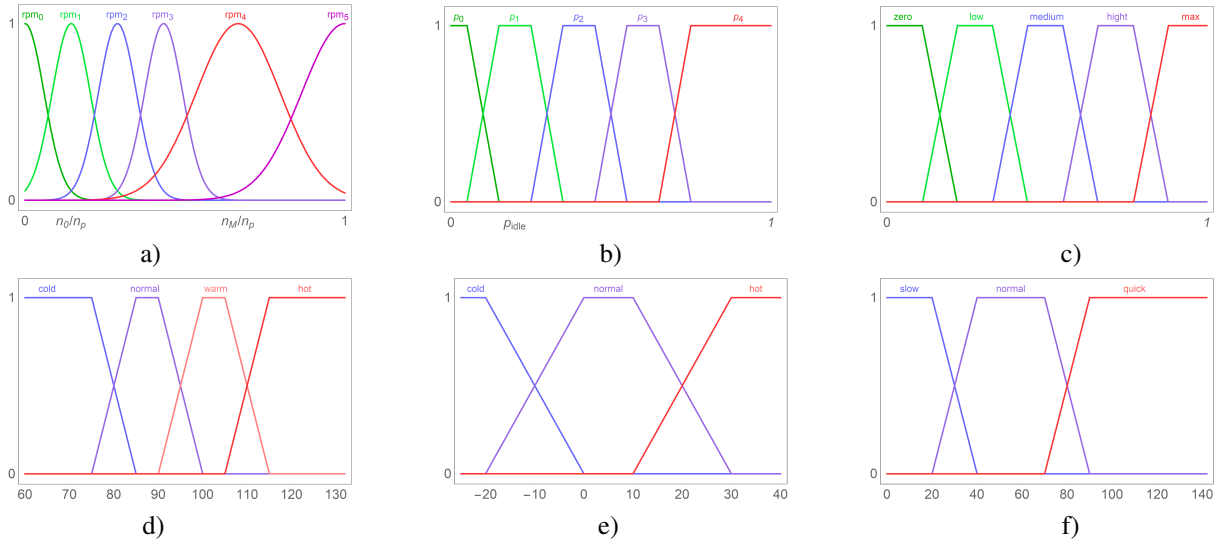


Fig. 7. Membership functions describing normalized fuzzy sets: rotational speed of the engine shaft a), negative pressure in the engine intake manifold b), torque developed by the engine c), engine temperature d), ambient temperature e), vehicle speed f).

acteristic values: idle speed, revolutions corresponding to the maximum torque and revolutions corresponding to the maximum engine power,

- negative pressure in the engine intake manifold - measured at the output of the absolute pressure sensor (MAP sensor), this parameter takes values from the range (0.1], where 1 corresponds to atmospheric pressure and 0 to vacuum, the intermediate characteristic value is the pressure for revolutions idling.

Each of these two quantities has been fuzzified and fuzzy rules have been defined to determine the motor load moment. Due to the fact that for different naturally aspirated gasoline engines, the characteristic points of rotational speed may be different and their proportions are similar, this parameter must be standardized for each car. Therefore the driver must enter the rotational speed for: idling, maximum torque and maximum engine power. These parameters are common for a given type of engine. Normalization is performed down to the $[0.1]$ range. Gaussian functions were used to blur the rotational speed of the motor shaft:

$$\mu_A(x) = \exp\left(\left(\frac{x-c}{2d}\right)^2\right). \tag{2}$$

Normalized membership functions describing the rotational speed of the motor shaft are shown in Fig. 7a). Due to the fact that the vacuum values in the engine intake manifold are in the range of $[0.1]$, this parameter does not need to be normalized. The blur was done using the trapezoidal membership functions. The fuzzy sets describing this parameter are presented in Fig. 7b). Based on a set of rules:

IF rpm is *R* **AND** pressure is *P* **THEN** moment is *M*

the knowledge base is included in Tab. I, in which the sets row for p_0 and the column for rpm_0 are omitted, because in both cases the answer is $zero$. Using the Zadeh model, we determine the load torque of the factorial, which is described by the normalized fuzzy sets shown in Fig. 7c). We used the

TABLE I
EXPERT KNOWLEDGE FOR MOTOR TORQUE LOAD CONTROL

		Crankshaft rotation per minute				
		rpm_1	rpm_2	rpm_3	rpm_4	rpm_5
Pressure	p_1	$zero$	$zero$	$zero$	low	$zero$
	p_2	$zero$	$zero$	low	$medium$	low
	p_3	$zero$	low	$medium$	$height$	$medium$
	p_4	$zero$	low	$height$	max	$height$

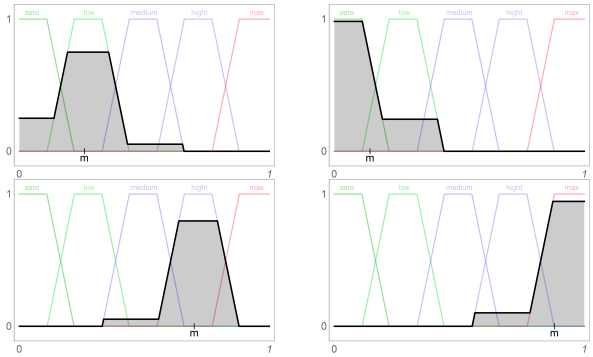


Fig. 8. Examples of determining the torque value based on rotational speed of the engine shaft and the vacuum in the intake manifold for the created fuzzy system.

formula for the center of mass for sharpening:

$$\text{output} = \frac{\int_X x \cdot m_{output}(x)dx}{\int_X m_{output}(x)dx}. \tag{3}$$

Sample examples of the system's responses are shown in Fig. 8, where m denotes the sharpened value of the torque load motors.

A. Fuzzy control for vehicle's driveline

The primary diagnostic system will determine the condition of the vehicle's driveline based on a fuzzy values:

- moment M, fuzzification is shown in Fig. 7c),

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TABLE II
EXPERT KNOWLEDGE FOR DEVELOPED CAR DIAGNOSTIC SYSTEM IN CASE OF ENGINE TEMPERATURE VALUE $T_e = warm$

$T_e = warm$		Moment											
		zero	low	medium			hight			max			
Velocity	low	*prob. bad*	*prob. bad*	Ambient temp.			Ambient temp.			*fine*			
				cold	normal	hot	cold	normal	hot				
				prob. bad	*prob. fine*	*fine*	*prob. bad*	*prob. fine*	*fine*				
	normal	*bad*	*bad*	*prob. bad*			*prob. fine*			*prob. fine*			
	quick	*bad*	*bad*	Torque converter clutch			Torque converter clutch			Torque converter clutch			
				off	on		off	on		off	on		
				prob. bad	*prob. fine*		*prob. bad*	*prob. fine*		*prob. bad*	*prob. fine*		

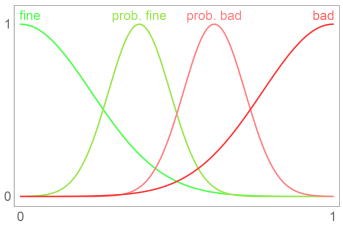


Fig. 9. Fuzzy sets describing primary diagnostic system membership functions to control condition of the vehicle drive system.

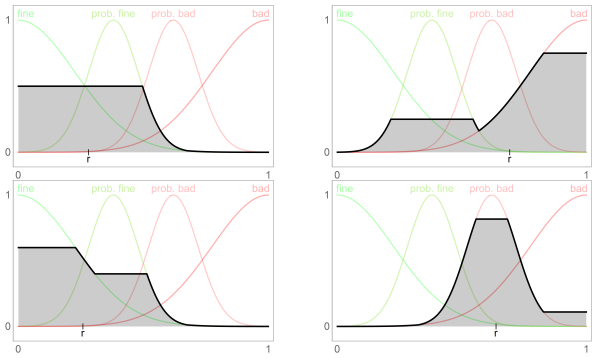


Fig. 10. Sample system responses for four different input states.

- engine temperature item T_e, fuzzification is shown in Fig. 7d),
- ambient temperature item T_a, fuzzification is shown in Fig. 7e),
- vehicle speed item V, where the fuzzification is shown in Fig. 7 e),
- state of torque converter clutch, as a singleton with two ON / OFF values.

The primary diagnostic system will detect possible excessive overheating of the engine under certain operation conditions. Due to the fact that the elements of the drive system have a certain thermal inertia, the system is based on the averaged values of the tested parameters in a fixed time - 5 minutes. We will use a moving average to determine the values of the input parameters. Based on the expert knowledge fuzzy control system was created using the following set of rules:

1) **IF** engine temperature is *cold* **THEN** machine condition is *probably bad*

2) **IF** engine temperature is *normal* **THEN** machine condition is *fine*

3) **IF** engine temperature is *warm* **AND** moment is *M* **AND** velocity is *V* **AND** ambient temperature is T_a **AND** TCC state is *S* **THEN** machine condition is *C*

4) **IF** engine temperature is *hot* **THEN** machine condition is *bad*

The rule 3) is described in a general way, where the knowledge is placed in Tab. II. In the table, in two cells for velocity *slow* and moment *medium* or *hight* the answer depends on the ambient temperature. This means that in other cases the ambient temperature has no influence on the form of the rules. For the measurement of the temperatures in and out of the radiator (T_2, T_3) and the temperature of the gearbox oil (T_6, T_7), fuzzy rules have been defined in a similar way, allowing the detection of insufficient cooling system efficiency. In this way, the system detected contamination of the air conditioning radiator, which reduced the efficiency of the engine radiator by restricting the air flow.

V. CONCLUSION

The fuzzy system proposed in the paper allows for early detection of faults, which may lead to damage to the car. As a composition of the IoT model was used trust mechanism which helps to improve knowledge extraction from participating cars. The system was tested in our project car. Results show that it is a valuable diagnostic tool, especially for owners of old vintage cars. First of all, it does not interfere with the construction of the car, it is cheap and provides valuable diagnostic information that is not provided by standard systems that a serial vehicle is equipped with.

In addition, the proposed trust mechanism is uncomplicated, as it requires calculation of similarity measure of fuzzy sets. Due to the fact that similarity measure is defined axiomatically, it is possible to define many different functions defining this measure, and this positively affects effectiveness of implemented mechanism.

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